Generative–Verifier Architecture for Regulation-Compliant Pharmaceutical Name Generation

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***Abstract*—Naming a medicine is a constrained search problem, not a creative one. A nonproprietary name must carry the mandated International Nonproprietary Name (INN) or United States Adopted Name (USAN) class stem, a proprietary name must carry none, and both must stay orthographically and phonetically distant from roughly twenty thousand marketed names whose confusion causes documented dispensing errors. This paper presents a generative–verifier architecture that couples two free-running statistical proposers to a five-check regulatory screen and a separate, downstream quality objective, so that generation and screening form one closed loop. The screen reimplements the algorithm family behind the FDA Phonetic and Orthographic Computer Analysis (POCA) tool, combining normalised Levenshtein similarity, BI-SIM bigram alignment and ALINE phonetic alignment over a rule-based grapheme-to-phoneme converter. Against 30 documented look-alike sound-alike pairs and 400 sampled random pairs it attains a ROC AUC of 0.997 with a mean score separation of 39.9 points. A three-seed sweep over twenty difficulty-stratified targets screens 8,081 candidates and admits 27.8%: *brand* targets, which carry no stem at all, accept at 0.501 ± 0.051 across seeds while stem-governed *generic* targets accept at 0.241 ± 0.003, the opposite objectives the two pipelines exist to enforce. Two results a winners-only evaluation would hide are reported: the 55–70 moderate-similarity band absorbs 88% of shortlisted names because a mandated stem itself forces similarity to siblings, and the highest-weighted quality term is a near-constant offset that does not change candidate ranking.**

***Index Terms*—pharmaceutical nomenclature, look-alike sound-alike drug names, phonetic similarity, constrained text generation, medication safety, regulatory screening**

I. INTRODUCTION

Confusion between drug names is a persistent and measurable source of medication error. Lambert *et al.* established that orthographic and phonetic similarity is a quantifiable risk factor for name-related dispensing errors rather than an anecdotal one [1], and the Institute for Safe Medication Practices maintains a standing list of name pairs implicated in real incidents [2]. Regulators respond to this at the point of approval: the US Food and Drug Administration screens every proposed proprietary name with the Phonetic and Orthographic Computer Analysis (POCA) tool [3] and publishes best-practice criteria that a sponsor is expected to have applied before submission [4], while the European Medicines Agency operates an equivalent review through its (Invented) Name Review Group [5].

The constraint set on a new name is therefore not aesthetic. A nonproprietary name is governed by the WHO INN stem system [6] and, in the United States, the USAN Council [7]: the name must terminate in the stem that identifies its pharmacological class, must not carry any *other* class's stem anywhere in the string, and must remain distinguishable from every sibling that shares the same mandated ending. A proprietary name is governed by the opposite rule. It must carry no class stem at all, since a stem in a brand name is a false claim of class membership, and it is additionally screened for trademark conflict and for adverse meaning in the markets where the product will be sold.

Existing computational work addresses only one half of this problem. The similarity-screening literature, beginning with Kondrak and Dorr's evaluation of orthographic and phonetic measures on confusable drug names [8], [9], treats the candidate name as given and asks how close it is to existing names. It does not propose names. Conversely, generative language models can produce plausible strings but have no notion of a mandated stem, a sibling set, or a regulatory operating point, and a model asked to produce a beta-blocker name will happily emit a real antibiotic prefix with the beta-blocker stem appended.

This paper describes a system that closes the loop. Its contributions are:

1) **A parallel pool-and-select architecture** in which two structurally different free proposers, a syllable grammar induced from the corpus and a guided character n-gram model, are both run on every request and the whole pool is screened before any selection is made, so a shortlist can never be a local optimum of whichever proposer happened to run first (Section III-E). Rejections return to the proposers as machine-readable payloads that steer the rest of the request.
2) **A five-check regulatory screen and a deliberately separate quality objective.** The screen reimplements

Project repository, full pipeline with reproducible experiments (v1.0): https://github.com/vedanshshetty/Pharmaceutical-Name-Generation/tree/main the POCA family from scratch and extends it with INN/USAN stem conformance (including detection of foreign class stems embedded in otherwise compliant names), a trademark proxy, phonotactics and a cross-lingual lexicon. The quality objective ranks what survives and can never rescue a rejection, keeping the regulatory decision a regulatory decision (Sections III-C and III-D).

3) **An evaluation that logs every candidate ever screened**, not only the winners, executed over three seeds and twenty targets with brand and generic results reported separately. Acceptance rate, per-proposer contribution, the failure-code distribution and an ablation of every objective term are all derivable from a single committed artifact (Section V).

Two findings are negative results about the system's own design and are presented as such: the review band regulators reserve absorbs most admissible names in a stem-governed class, and the highest-weighted term in the quality objective shifts every score by a near-constant amount without changing which names are selected.

II. Background and Related Work

A. Regulatory constraints on pharmaceutical names

The INN system assigns a common stem to substances sharing a pharmacological class, so that `-olol` identifies a beta-adrenoceptor antagonist and `-prazole` a proton pump inhibitor [6]. The stem is mandatory and terminal for a nonproprietary name; the portion preceding it, conventionally called the fantasy prefix, is the only part the namer controls. USAN operates a parallel and largely harmonised list for the United States [7]. The FDA's best-practice guidance for proprietary names additionally prohibits names that overstate efficacy, imply an unapproved use, or are confusable with a marketed product [4], and the EMA guideline imposes comparable conditions for centrally authorised products [5].

B. Computational similarity screening

Kondrak and Dorr evaluated a large family of string and phonetic similarity measures against a reference set of confusable drug name pairs and reported that combining an orthographic bigram measure, BI-SIM, with edit distance outperformed any single measure [8], [9]. BI-SIM aligns bigram sequences with a longest-common-subsequence recurrence in which a pair of bigrams may match partially, recovering the transposition-style confusions that unit-cost edit distance [10] under-penalises. On the phonetic side, ALINE [11] scores an optimal alignment between two phone sequences using articulatory feature distances. Zobel and Dart established the broader point that phonetic string matching benefits from combining orthographic and phonetic evidence rather than choosing between them [12]. Emmerton *et al.* reported an operational LASA-detection tool on similar foundations [13], and Lambert *et al.* characterised detection limits for identifying highly confusable names from experimental data [14]. The FDA's POCA tool implements a comparable composite and is the practical operating reference [3].

C. Generative approaches

Constrained decoding can force a language model's output to satisfy a formal grammar without finetuning [15], and controlled sampling schemes such as nucleus sampling govern the quality-diversity tradeoff in open-ended generation [16]. Surveys of large language models in pharmaceutical research and development document rapid adoption alongside a persistent tendency to generate confident but unverifiable output [17]. Neither line of work supplies what pharmaceutical naming needs, which is not fluency but a decision procedure defensible to a regulator.

D. The gap this work addresses

Screening systems evaluate names they are given. Generative systems produce names nobody has screened. The proposed system treats the two as one closed loop in which the screen's structured rejections are fed back as sampling pressure within the same request, and in which admissibility and desirability are computed by two deliberately separate components.

III. System Architecture

A. Reference data and corpus construction

The data layer assembles a screening universe from primary regulatory sources: the openFDA National Drug Code directory [19], the RxNorm ingredient and brand-name vocabularies [18], and the EMA medicines register, plus a curated INN/USAN stem table of 277 entries. It is live-first with a committed static snapshot as fallback, and every run records which path executed, because degrading silently to a smaller universe would inflate every distinctiveness margin reported.

The raw feed is not usable as a name corpus. The NDC directory mixes true drug names with over-the-counter product descriptions, so a filter admits only single alphabetic tokens of 4 to 25 characters that are not dosage forms, salts or counterions, or marketing modifiers. Multi-ingredient generic entries are split on whitespace rather than discarded, because each component of a combination product is itself a real active-ingredient name. Table I and Fig. 2 report the result; the surviving names have a mean length of 9.7 characters and 4.0 syllables for generics against 7.8 and 3.1 for brands, and these empirical distributions are the reference used by the shape term of the objective.

B. Proposers

1) Induced syllable grammar: The grammar proposer samples a fantasy prefix from a syllable inventory *induced* from the corpus rather than hand-specified. Real prefixes are parsed into onset, nucleus and coda units, and new syllables are drawn from the empirical positional frequencies (Fig. 3). Induction over the 1 700 extracted prefixes yields 39 attested onsets for the generic register and 56 for the

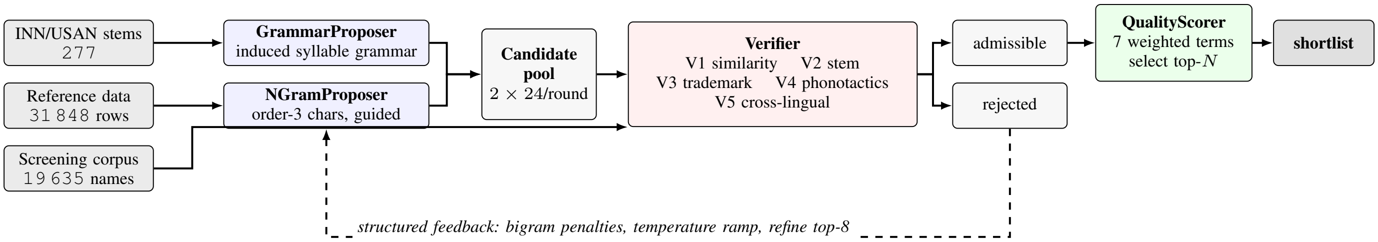

Fig. 1. Request path of the proposed system. Both free proposers run in parallel on every request and the entire pool is screened before any selection is made, so a shortlist can never be a local optimum of whichever proposer ran first. Rejections return to the statistical proposer as machine-readable payloads, never as prose.

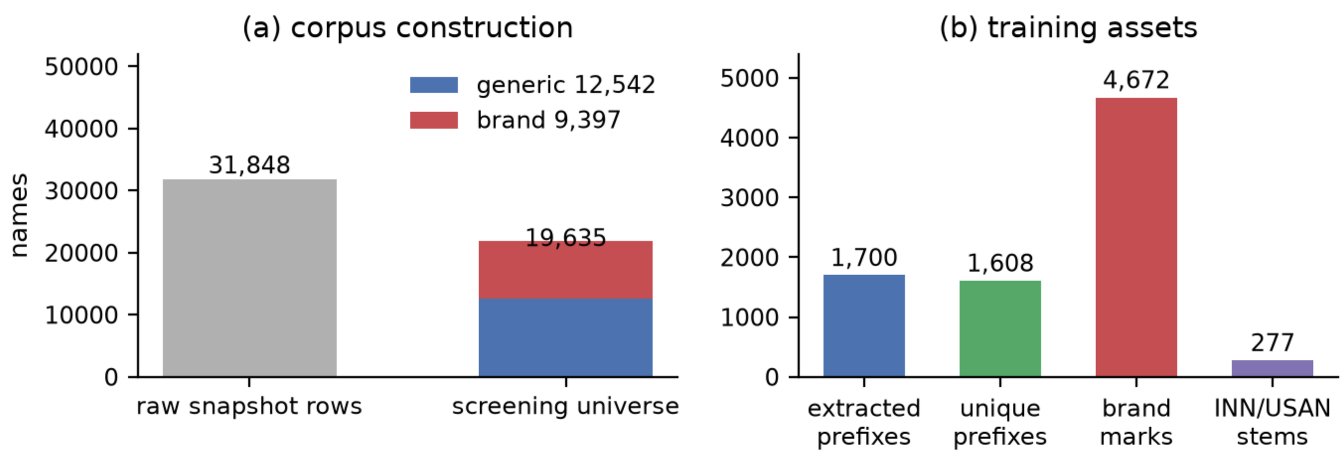

Fig. 2. Corpus statistics drawn by `paper/make_figures.py` from the committed manifest. (a) Of 31 848 raw snapshot rows, filtering admits a 19 635-name screening universe split 12 542 generic / 9 397 brand. (b) The training corpora the two proposers are fit on: 1 608 unique fantasy prefixes, a 4 672-mark brand set, and the 277-row INN/USAN stem table.

TABLE I
CORPUS COMPOSITION OF THE REPORTED RUN.

Quantity	Value	Note
Raw snapshot rows	31 848	multi-source union
Screening universe (unique)	19 635	what the verifier measures from
generic names	12 542	
brand names	9 397	trademark screen pool: 9 402
Noise tokens filtered	48 799	dosage forms, salts, marketing
Fantasy prefixes extracted	1 700	1 608 unique; n-gram training set
Stem matches in corpus	1 817	14.5% of generic tokens
Brand marks (training)	4 672	
Blocklisted morphemes	1 212	class-signalling fragments
INN/USAN stems	277	suffix and embedded matching

brand register, with an empirical syllable-count distribution of $\{1{:}233,\ 2{:}846,\ 3{:}517,\ 4{:}99,\ 5{:}4\}$.

The design consequence is structural rather than statistical. Because recombination happens at the syllable level, the proposer cannot reproduce a whole morpheme such as `erythro` unless `e`, `ry` and `thro` are independently resampled in order, so the memorisation failure mode is excluded by the representation rather than penalised after the fact. When the mandated stem begins with a vowel the sampler is further constrained to emit a consonant-final prefix, avoiding a vowel pile-up at the join.

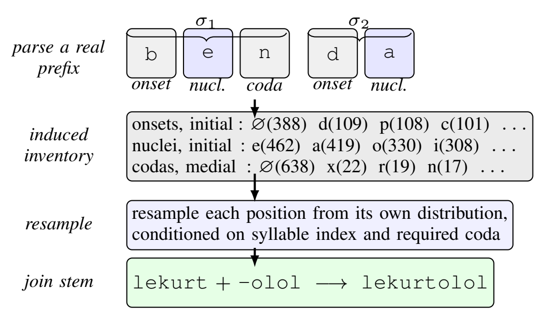

Fig. 3. Syllable-grammar induction on the real corpus prefix `benda`. Counts are observed positional frequencies over the 1 700 extracted prefixes; ∅ is an empty onset or coda. Onsets are maximised, so an intervocalic consonant becomes a coda only when the remainder is not an attested word-initial onset.

2) Guided character n-gram model: The statistical proposer is an order-3 character model with add-k smoothing trained on the same prefix set. Guidance operates across a batch rather than within a draw (Fig. 4). Bigrams appearing in the colliding region of a rejected candidate are down-weighted but never zeroed, so the search space stays connected, and the sampling temperature rises with the number of consecutive failed rounds. Each output slot is drawn three times and the draw with the highest verifier-free reward is kept, which approximates

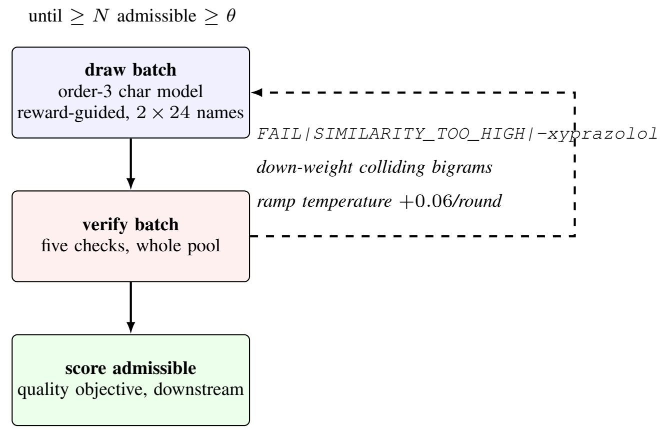

Fig. 4. Guided n-gram proposal loop. Bigrams from rejection payloads steer later draws in the same request; admissible names alone proceed to the quality objective. Everything that fails a check is logged, never silently dropped.

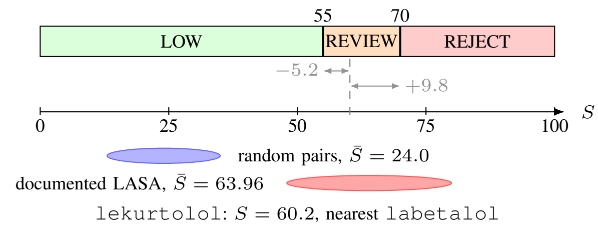

Fig. 5. The two-cutoff decision structure, drawn to scale, with the measured mean composite scores of the 400 random and 30 documented look-alike sound-alike pairs superimposed. A candidate reports *two* margins. Reporting only the margin to 70, as an earlier version did, makes a name inside the review band read as comfortable.

optimising for quality during proposal at microsecond rather than millisecond cost.

C. The screening component

The verifier answers one question, whether a name *may* exist, through five checks. It is a from-scratch implementation with no model downloads and no network calls on the default path, which is what makes a results run byte-reproducible.

a) V1, similarity: For candidate a and corpus name b, orthographic similarity combines normalised Levenshtein similarity L with BI-SIM B, and the composite blends that with ALINE phonetic similarity P computed over rule-based phoneme transcriptions:

$$L(a,b) = 1 - \frac{d_{\text{lev}}(a,b)}{\max(|a|,|b|)}, \tag{1}$$

$$O(a,b) = w_L\, L(a,b) + w_B\, B(a,b), \tag{2}$$

$$S(a,b) = 100 \cdot \left[\, w_O\, O(a,b) + w_P\, P(a,b) \,\right], \tag{3}$$

with $w_L = w_B = w_O = w_P = 0.5$, reproducing the equal-weight configuration of the published POCA behaviour. The composite risk score of a candidate is $\max_b S(a,b)$ over the screening universe, computed after a bigram prefilter that reduces the phonetic comparison to a pool of 150 nearest candidates.

Two operating points partition the score, following the published reading of POCA output: $S \geq 70$ is a hard rejection, $55 \leq S < 70$ is a moderate-similarity band admissible with review, and $S < 55$ is low risk (Fig. 5). When the target class mandates a stem, the stem is stripped from both strings before scoring siblings that share it, so that distinctiveness measures the part the namer actually controls rather than penalising a name for complying with the regulation that forced the ending on it. Sibling comparison uses a separate and stricter cutoff of 75.

b) V2, stem conformance: For a generic target the terminal stem must match the requested class, at least two characters of fantasy prefix must precede it, and no *foreign* class stem may appear embedded elsewhere in the string. The last condition is not redundant. Names such as `cillinolol` carry a correct terminal stem and a contradictory internal one, and the INN system treats a misleading internal stem the same way it treats a misleading terminal one. For a brand target the condition inverts: any stem in stem position is a false class claim and fails.

c) V3, trademark collision: The candidate is scored against the 9 402 marketed proprietary names using the same composite, with a rejection at 70. This is explicitly a proxy for marketed-name collision, not a registered trademark class search.

d) V4, phonotactic well-formedness: The name is transcribed by a rule-based grapheme-to-phoneme converter, syllabified by sonority, and scored as $0.6\,\phi + 0.4\,\tau$ where ϕ is phonotactic legality (cluster admissibility, consonant and vowel run limits) and τ is corpus typicality under a character trigram model of the appropriate register. A name yielding no vowel nucleus is rejected outright; the floor is 0.45.

e) V5, cross-lingual and implied claim: A curated lexicon of 68 adverse terms across eight major markets (Spanish, French, German, Italian, Portuguese, Japanese, Chinese, Hindi) is matched as substring and by ALINE similarity at a 0.88 cutoff, and 25 efficacy-implying fragments are flagged as warnings. This is a lexicon lookup, not a phonological model.

D. The quality objective

The verifier decides admissibility; it says nothing about whether a name is any good. That distinction matters because an earlier version of this system, measuring risk alone, accepted `erythroolol`, `amoxiolol` and `acycloolol`: three real drug prefixes with a beta-blocker stem appended, which is exactly the cross-class misidentification the stem system exists to prevent.

A separate scorer therefore computes seven weighted terms in $[0, 1]$, with two profiles because the objectives genuinely differ (Table II). Two choices are worth stating. *Distinctiveness* scores headroom below the review line at 55, not the reject line at 70, so a grey-band name scores poorly rather than comfortably. *Typicality* is scored as a band targeting 0.40 to 0.70 rather than a maximum, because maximising corpus likelihood is precisely the objective that produces memorised prefixes.

TABLE II

QUALITY OBJECTIVE. WEIGHTS SUM TO 1.0 WITHIN EACH PROFILE; A DASH MARKS A TERM THAT DOES NOT EXIST FOR THAT PROFILE.

Term	Gen.	Brand	Measures
distinctiveness	0.24	0.24	headroom below the review line
novelty	0.20	0.14	verbatim reuse of a real fragment
morpheme hygiene	0.16	–	carrying another class's stem
stem avoidance	–	0.16	any stem is a false class claim
memorability	–	0.15	short, few syllables, clean CV
pronounceability	0.14	0.14	phonotactic legality, ease
shape	0.10	–	length, syllables vs. real names
seam	0.09	0.09	hygiene at the prefix/stem join
typicality	0.07	0.08	plausible without being derivative

Algorithm 1 Generative–verifier pool-and-select generation

Require: target type t, class c, stem s, shortlist size N, seed
$\mathcal{E} \leftarrow \emptyset$; $\mathcal{A} \leftarrow \emptyset$ ▷ all screened; admissible
for round $r = 1 \ldots R_{\max}$ **do**
  $\mathcal{P} \leftarrow \bigcup_{p \in \{\text{grammar,ngram}\}} p.\textsc{Propose}(24, \text{ctx})$
  for all $n \in \mathcal{P} \setminus \textsc{Seen}$ **do**
    $\rho \leftarrow \textsc{Verify}(n, t, c, s)$; $q \leftarrow \textsc{Quality}(n, \rho, t, s)$
    $\mathcal{E} \leftarrow \mathcal{E} \cup \{(n, \rho, q)\}$
    if $\rho.pass$ **then** $\mathcal{A} \leftarrow \mathcal{A} \cup \{(n, \rho, q)\}$
    else LEARNFROMREJECTION(ctx, n, ρ)
    end if
  end for
  for all $n \in \textsc{TopK}(\mathcal{P} \setminus \mathcal{A}, 8)$ by q **do**
    refine n for up to 3 rounds; screen and score each edit
  end for
  if $|\{a \in \mathcal{A} : q_a \geq \theta\}| \geq N$ **then break**
  end if
end for
return top-N of $\mathcal{A}$ by quality, and $\mathcal{E}$ as the audit log

Critically, the scorer is never consulted by the admissibility decision. It ranks what survives and cannot rescue a rejection, which is what keeps the regulatory claim a regulatory claim rather than a preference model.

E. Pool-and-select orchestration

Algorithm 1 states the request path. Nothing is discarded before it has been screened. An early-exit cascade optimises for cost: if the first proposer clears the bar with a margin of 1.04 it stops, and never learns that the other had a margin of 15 on the same call. Pooling screens everything before choosing, so selection cannot be a local optimum of proposer ordering. Sampling stops only when at least N admissible candidates clear a configured quality target θ; there is no separate escalation stage to skip.

Rejections drive two mechanisms. Bigrams drawn from the machine-readable payload of each rejection signal, specifically the colliding name, the offending sibling and the foreign morpheme, become sampling penalties for the rest of the request. Separately, the eight highest-quality rejects are sent through up to three targeted edit rounds, because editing a candidate that scored 61 and missed on one code is the cheapest quality in the system, while editing one that scored 20 is wasted work.

IV. EXPERIMENTAL SETUP

The open-source implementation `pharma_name_gen` is built by a single builder used by the notebook, the web application and the sweep harness, so no entry point can silently run a different configuration. All numbers below are reproduced offline from the committed data snapshot (fingerprint `d9cc79de5be14f36`), by `paper/reproduce.py`, with no network access and no API keys. The reported figures are not configuration-specific artefacts: executed against *live* openFDA and RxNorm feeds, the discrimination experiment returns the identical corpus fingerprint and identical statistics on the same 30 positive and 400 negative pairs.

The pipeline configuration is 24 candidates per proposer per round, at most 4 rounds, shortlist size 10, up to 3 refinement rounds over the top 8 rejects, temperature 1.0 with a ramp of 0.06 per failed round, 3 reward-guided draws per slot, and a quality target $\theta = 62.0$. The verifier configuration is the threshold set given in Section III-C with stem-aware similarity enabled, top-k of 5, a prefilter pool of 600 and a phonetic pool of 150.

Reproducibility is enforced structurally. Trained artefacts are content-addressed on the corpus fingerprint plus a configuration hash, so changing the stem table produces a different key and a stale model is simply not found. Every run emits a manifest recording the fingerprint, data mode, source record counts, thresholds, weights and git commit. A suite of 35 tests, including an assertion that seeded runs are byte-identical, passes offline.

The sweep spans twenty targets chosen to cover the difficulty range rather than sampled at random, since a sweep over easy classes only would report a flattering acceptance rate that says nothing. Classes are stratified as *roomy* (few siblings, short names viable), *crowded* (many siblings, high intra-stem pressure), *saturated* (dozens of recent entrants, long stems), and *brand* (no stem, inverted objective). Because the brand register is the opposite objective rather than an easier generic class, the brand cell set was widened from two to six therapeutic contexts (cardiovascular, oncology, CNS/neurology, anti-infective, respiratory, dermatology) so that the brand/generic comparison rests on six cells, not two. The whole sweep is run at three seeds, $\{1, 2, 3\}$, so the headline rates are reported as mean $\pm$ standard deviation across seeds rather than as a single point estimate; a full sweep screens roughly 2 700 candidates and runs in about twenty minutes on a consumer CPU.

V. RESULTS

A. Does the screen discriminate?

The screening component is validated independently of anything the generator produces. Positives are 30 pairs from the FDA Name Differentiation Project and the ISMP list of confused drug names [2], a far stronger label than visual resemblance because each pair has caused documented dispensing errors. Negatives are 400 pairs sampled uniformly from the screening corpus (Table III, Figs. 6 and 7).

The finding worth stating is not the AUC but the shape of Fig. 7: specificity saturates at 1.000 by the review line, so

TABLE III
DISCRIMINATION OF THE COMPOSITE SIMILARITY SCORE.

Quantity	Value
Documented confusable pairs, n	30
Random corpus pairs, n	400
Mean composite, confusable	63.96
Mean composite, random	24.04
Separation	39.91
ROC AUC	0.9970
Sensitivity / specificity at 55 (review)	0.767 / 1.000
Sensitivity / specificity at 70 (reject)	0.367 / 1.000
Youden-optimal threshold ($J = 0.952$)	40.0
Sensitivity / specificity at 40	0.967 / 0.985

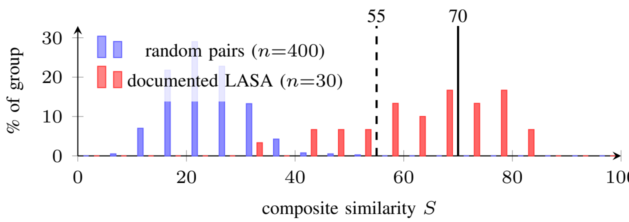

Fig. 6. Score distributions for documented confusable pairs and random corpus pairs, binned at width 5 and normalised within each group so that the two very different sample sizes are comparable. The distributions are almost disjoint, which is what a ROC AUC of 0.997 means concretely, and the residual overlap sits entirely below the review line.

every threshold point above 55 is paid for purely in sensitivity. At the reject cutoff of 70 the screen recovers only 11 of the 30 documented confusable pairs. A system reporting a binary pass at 70 would silently discard most of its own detection capability; the two-cutoff design is what preserves it.

B. Which similarity measure carries the discrimination?

Each weight was zeroed in turn and the AUC recomputed against 300 sampled negatives (Table IV). The orthographic component is load-bearing; removing the phonetic and BI-SIM terms changes the AUC by less than 0.001. The methodology must reflect this: the claim that phonetic alignment contributes is not supported at this operating point by these 30 pairs, though it is retained because the pairs it targets, homophones with divergent spellings, are under-represented in a set of 30.

C. Does pool-and-select beat independent strategies?

The predecessor architecture exposed four peer generation strategies, each running its own end-to-end loop and producing the whole shortlist alone. Its verbatim output on the `-olol` class is scored here by the *same* quality function against the *same* corpus, which is the only way to turn the comparison into a number rather than an assertion, since the earlier system had no quality function at all. Table V and Fig. 8 report the comparison.

Two details in Fig. 8 matter more than the mean. The predecessor's best name, `hiemfailolol`, outscores every pool-and-select candidate, so the improvement is not a higher ceiling but a higher floor: v2's worst candidate scores 64.55 against a v1 minimum of 27.24, and the variance collapses.

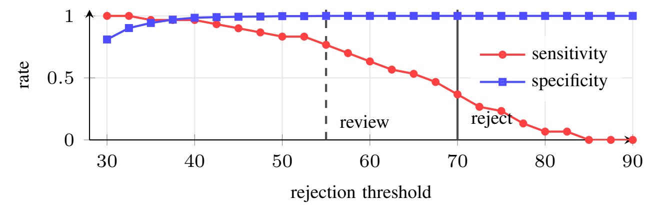

Fig. 7. Operating characteristics. Specificity against random pairs is already 1.000 at the review line of 55, so raising the cutoff to 70 buys no additional specificity and costs 0.40 in sensitivity. This is the direct quantitative argument for treating 55–70 as a reportable band rather than passing it silently.

TABLE IV
SIMILARITY-COMPONENT ABLATION. ONE WEIGHT ZEROED AT A TIME, 30 POSITIVE AND 300 NEGATIVE PAIRS.

Component zeroed	Weight	ROC AUC	Δ
(none removed)	–	0.9974	0.0000
$w_{\text{orthographic}}$	0.5	0.9892	-0.0082
$w_{\text{levenshtein}}$	0.5	0.9963	-0.0011
w_{phonetic}	0.5	0.9969	-0.0005
$w_{\text{bi-sim}}$	0.5	0.9974	0.0001

The three worst v1 names, `erythroolol`, `vancicloolol` and `acycloolol`, are all real drug prefixes with a beta-blocker stem appended, the failure mode the induced grammar structurally excludes and the morpheme-hygiene term penalises; that term rises from 0.879 to 1.000.

Reproducibility was verified directly: three independent invocations with the same seed returned byte-identical shortlists.

D. Brand versus generic across three seeds

The sweep logs one row per candidate ever screened, accepted or not. Across twenty targets and three seeds it screened 8 081 candidates and admitted 2 247 of them (27.8%); each seed screened roughly 2 700 candidates and each returned a full shortlist of 10 for every target, entirely on CPU.

The comparison the two pipelines exist to make is reported separately (Fig. 9). *Brand* targets carry no stem and therefore no sibling set, and they accept at 0.501 ± 0.051 across seeds; *generic* targets, to which final verification attaches twice, accept at 0.241 ± 0.003. The spread between seeds is one order of magnitude larger for brand (± 0.051) than for generic (± 0.003), for a structural reason: a stem-governed shortlist is anchored by the mandated ending, so its accept rate is dominated by the corpus, while a brand shortlist is what the proposers happen to draw in a given seed. Generic shortlists are also the better ones: best generic quality is 75.4 ± 0.2 against 68.2 ± 1.0 for brand, because a short distinctive brand mark must by construction rest closer to the marketed brand corpus (mean composite risk 70.2, against 63–66 for stem-governed classes).

The stratification predicts. Pooled by stratum, the acceptance rate over stem-governed classes falls monotonically from roomy (1 356 candidates, 0.308, mean q 63.7) through crowded (2 972, 0.225, 62.4) to saturated (2 601, 0.225, 61.5), and the effort rises with it: 3.3 candidates screened per admitted name

TABLE V
ARCHITECTURE COMPARISON ON `-olol`, SAME CORPUS, SAME SCREEN, SAME SEED. PER-COMPONENT FIGURES ARE MEANS OVER EACH ARCHITECTURE'S OUTPUT.

	v1: four strategies	v2: pool-and-select
Candidates returned	15	10
Mean quality	55.38	**67.33**
Best quality	72.63	68.98
Mean composite risk	59.31	59.09
Low-band names	4	1
Mean per-component quality		
novelty	0.560	**0.928**
seam	0.737	**1.000**
morpheme hygiene	0.879	**1.000**
pronounceability	0.936	**0.979**
shape	0.819	**0.921**
typicality	0.179	0.118
distinctiveness	0.040	0.001

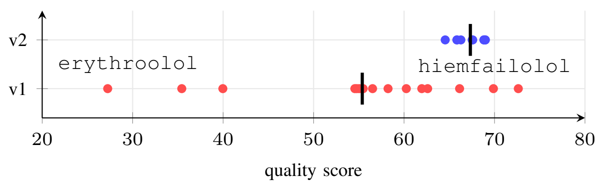

Fig. 8. Per-candidate quality for the two architectures on the same class, corpus and seed; vertical bars mark the means. The predecessor's spread is the point: one candidate above every v2 name and three below 40.

TABLE VI
PER-CLASS SWEEP RESULTS MERGED OVER THREE SEEDS, SHORTLIST SIZE 10 IN ALL CASES. THE LOW/MOD COLUMN COUNTS RISK BANDS OVER THE ADMITTED CANDIDATES.

Stem	Difficulty	Scr.	Adm.	Rate	Best q	low/mod
`-pril`	roomy	196	100	0.510	71.27	5/95
`-sartan`	roomy	547	113	0.207	70.62	10/103
`-vaptan`	roomy	202	101	0.500	73.15	9/92
`-dipine`	roomy	411	103	0.251	72.83	4/99
`-olol`	crowded	194	114	0.588	71.69	13/101
`-prazole`	crowded	824	150	0.182	68.60	10/140
`-caine`	crowded	413	90	0.218	69.18	2/88
`-statin`	crowded	809	167	0.206	67.37	0/167
`-cycline`	crowded	732	148	0.202	68.18	16/132
`-tinib`	saturated	623	124	0.199	71.45	2/122
`-mab`	saturated	180	133	0.739	69.52	3/130
`-gliptin`	saturated	814	119	0.146	66.51	21/98
`-gliflozin`	saturated	799	90	0.113	61.14	62/28
`-parib`	saturated	185	119	0.643	72.45	6/113
brand (all six)	brand	1152	576	0.500	67.03	24/552

TABLE VII
FAILURE PROFILE OVER 5 834 REJECTED CANDIDATES (THREE SEEDS).

Failure code	Count	Share
`TRADEMARK_HIT`	5324	0.913
`INTRA_STEM_TOO_CLOSE`	3314	0.568
`SIMILARITY_TOO_HIGH`	1899	0.326
`EXACT_NAME_COLLISION`	17	0.003
`STEM_MISUSE_IN_BRAND`	6	0.001
`STEM_FOREIGN_EMBEDDED`	4	0.001
`CROSSLINGUAL_ADVERSE_MEANING`	1	0.000

in roomy classes against 4.4 in crowded and saturated ones, while brand needs 2.0. Per-class detail (Table VI) shows a wide within-stratum spread: `-mab` admits at 0.739 despite being nominally saturated, because the stem is short and its sibling set orthographically diverse, while `-gliflozin` admits at 0.113 because a nine-character mandated stem leaves very little of the name under the namer's control.

E. Which checks do the rejecting?

A check that never fires is either redundant or misthresholded, and either way its presence in a methodology section is not supported by the data. Table VII gives the distribution of failure codes over the 5 834 rejections; the shares exceed 1.0 in total because a candidate can fail several checks at once.

Three checks account for essentially all rejections and the remaining four are close to inert, which is not evidence that they are unnecessary. The phonotactic check did not fire once across all three seeds, because the induced grammar makes unpronounceable output structurally rare, so it is confirming that the proposer works, and `STEM_FOREIGN_EMBEDDED` fires on names that would otherwise have passed with a correct terminal stem and a contradictory internal one. The trademark and intra-stem checks are the binding constraints; the others are cheap insurance.

F. Proposer contribution

Both free proposers were run on every request, so their contributions can be compared over logged pool data rather than by forcing a mode switch on the user (Table VIII).

The two have near-identical acceptance rates, 0.266 against 0.292, and the n-gram model produces the single highest-quality candidate of the sweep. Yet the grammar proposer takes 69.2% of shortlist slots, because acceptance and quality are different questions and selection happens on the latter. This split is the empirical case for pooling: a cascade running the n-gram model first and stopping on its first admissible output would have been right about admissibility and still produced a materially worse shortlist.

Refinement contributes disproportionately relative to its own acceptance rate. Of 2 296 refined attempts only 9.8% were admitted, against 35.0% for unrefined candidates, but refined candidates carry a mean quality of 65.41 against 60.80 and occupy 137 of the 600 shortlist slots. Editing near-misses is a low-yield, high-value operation, which is why it is restricted to the top 8 rejects.

G. Does every quality term earn its place?

Each term was removed in turn and the composite recomputed over the 6 929 generic candidates in the sweep, and the result reported as a Spearman rank correlation against the full objective, because what matters is whether removal changes *which names get picked*, not whether it shifts every score by a constant (Table IX).

The result is uncomfortable and is reported as such. The *highest*-weighted term, distinctiveness at 0.24, shifts absolute scores by 19.58 points on average yet leaves the ranking almost untouched at a rank correlation of 0.999. The reason is in

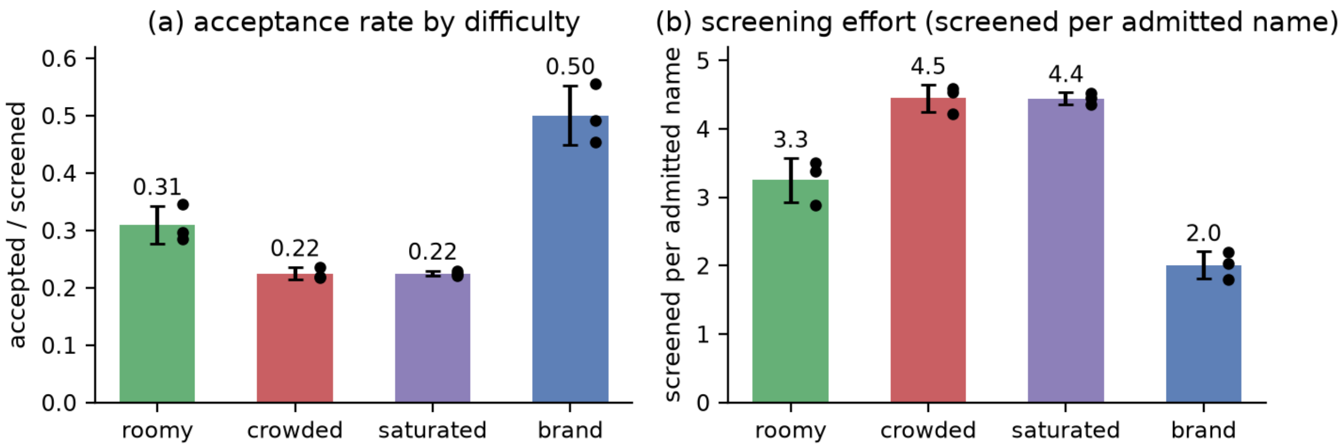

Fig. 9. Acceptance rate and screening effort by difficulty stratum across the three seeds (bars: mean; whiskers: standard deviation; dots: per-seed values). The seed spread is an order of magnitude larger for brand than for generic, because brand shortlists are anchored by what the proposers draw in a given seed while stem-governed ones are anchored by the mandated ending itself.

TABLE VIII
PROPOSER CONTRIBUTION OVER THE FULL THREE-SEED SWEEP.

Proposer	Prop.	Acc. rate	Mean q	Best q	Shortlist
grammar	4251	0.266	63.60	75.33	0.692
`ngram_guided`	3830	0.292	60.47	75.65	0.308

TABLE IX
QUALITY-OBJECTIVE ABLATION, $n = 6\,929$ GENERIC CANDIDATES (THREE SEEDS).

Term removed	w	Rank corr.	$\lvert\Delta q\rvert$	Reading
novelty	0.20	0.545	5.06	load-bearing
typicality	0.07	0.938	3.54	contributes
pronounceability	0.14	0.946	3.98	contributes
shape	0.10	0.950	2.70	contributes
morpheme hygiene	0.16	0.991	6.79	near-inert
seam	0.09	0.998	3.66	near-inert
distinctiveness	0.24	0.999	19.58	near-inert

TABLE X
AUDIT TRAIL FOR AN ADMITTED AND A REJECTED CANDIDATE, `-olol`.

	`lekurtolol`	`erythroolol`
Admissible	yes	**no**
Risk band	moderate	high
Composite risk S	60.21	76.15
Nearest existing name	`labetalol`	`erythrulose`
orthographic / phonetic	47.2 / 73.2	69.3 / 83.0
Margin to review (55)	-5.21	-21.15
Margin to reject (70)	$+9.79$	-6.15
Syllables	4	5
Phonotactic / typicality	1.00 / 0.011	1.00 / 0.176
Quality total	**68.98**	**27.24**
novelty	1.000	0.150
morpheme hygiene	1.000	0.120
seam	1.000	0.350
shape	0.998	0.821

the raw data: mean distinctiveness across all screened generic candidates is 0.003 with a standard deviation of 0.021. The term is saturated at zero, because it scores headroom below the 55 review line and virtually every stem-conforming candidate lands above that line, so it contributes a near-constant offset rather than a discriminating signal. Morpheme hygiene is near-inert for the opposite reason, with a mean of 0.979 because the induced grammar makes foreign morphemes structurally rare. Only novelty genuinely reorders the shortlist, at 0.545.

VI. Discussion

A. The review band is a property of the problem

Of the 600 names returned across the sweep, 528 (88.0%) sit in the 55–70 review band and only 72 clear it outright. Rejecting the band outright was tried and collapsed the admissible rate to roughly 0.5%. The explanation is structural rather than algorithmic: in a stem-governed class the mandated stem is shared verbatim with every sibling, so a name complying with the regulation is thereby forced into high orthographic similarity with the entire class. Stem-aware scoring mitigates but does not eliminate this, because comparison against the rest of the corpus still uses the full string.

The correct response is to report the band explicitly and let the quality objective rank grey-band names below clear ones, which is why the objective scores headroom against 55 rather than 70. Combined with Table IX, however, the two facts compose awkwardly: distinctiveness measured against 55 is exactly the term that saturates, so the mechanism intended to demote grey-band names is the one with the least effect on ranking. Rescaling it against the conditional score distribution within a class is the obvious correction and is left to future work.

B. What the audit trail looks like

Table X contrasts a name the system returned with one its predecessor returned, using the same screen; the converter transcribes `lekurtolol` as `L EH K ER T OW L AO L`.

`erythroolol` is rejected on similarity to `erythrulose` at 76.15 and independently scores 0.150 on novelty and 0.120 on morpheme hygiene because `erythro` is a real class-signalling morpheme: two signals pointing at the same defect from different directions, which is the intended redundancy.

VII. Limitations

This is a first-pass screen. It contains no prescription-simulation human-factors study, no legal likelihood-of-confusion analysis and no committee judgement, all part of an actual regulatory review [4], [5]. A low band means *worth taking to review*, never *cleared*.

The universe bounds the claim. Distinctiveness is measured against the assembled corpus of 19 635 names, so every reported margin is an upper bound. The EMA feed failed to retrieve in the runs recorded here, leaving European-only products represented solely through the committed snapshot.

Phonetics are orthographic and Anglocentric. The grapheme-to-phoneme converter is rule-based and English-oriented, and the cross-lingual check is a 68-term lexicon lookup, not a phonological model of each market language.

The trademark check is a proxy. It screens against marketed proprietary names, not against a registered trademark class search, which is a legal instrument and not reproducible from open data.

The validation set is small. Thirty documented confusable pairs is a strong label but a thin sample, and the component ablation in Table IV is underpowered as a result: a difference of 0.0005 in AUC over 30 positives is not a meaningful effect size.

Residual artefacts survive selection. The `-olol` shortlist includes `ipqolol`, whose `pq` sequence passes the phonotactic check on the transcribed phoneme string but reads poorly: the check tests syllable-level legality and does not penalise low-frequency orthographic sequences directly.

VIII. Conclusion and Future Work

This paper describes a generative–verifier architecture that treats pharmaceutical naming as a constrained search problem in which generation and regulatory screening form one closed loop rather than two disconnected tools. The screen discriminates documented confusable pairs from random corpus pairs at a ROC AUC of 0.997 with a mean separation of 39.9 points, and the generator returns a full shortlist for all twenty targets of a difficulty-stratified sweep at three seeds, admitting 27.8% of 8 081 screened candidates entirely on CPU with no API dependency. The two pipelines deliver the opposed behaviours their regulatory mandates demand: generic targets accept at 0.241 while brand targets accept at 0.501, and the best generic shortlist quality exceeds the best brand one.

Four results point at future work. The moderate-similarity band absorbs 88% of returned names because a mandated stem forces similarity to siblings, so the distinctiveness term should be rescaled against the conditional score distribution within a class rather than a fixed line. The similarity-component ablation is underpowered at 30 labelled pairs; extending the ground truth to the full published ISMP list with severity weighting would test the phonetic component properly. The near-inert highest-weighted term is a standing invitation to re-derive the objective's weights from data rather than by hand. Finally, replacing the Anglocentric rule-based converter with a multilingual grapheme-to-phoneme model would turn the cross-lingual check into a phonological screen rather than a lexicon lookup.

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